

Machine Learning for Natural Language Understanding

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A wide-angle photograph of the Stonehenge monument in England. The large, grey, moss-covered stone structures are arranged in their characteristic circular formation on a green grassy field. Several people are visible walking around the base of the stones, providing a sense of scale. The sky is a clear, pale blue with a few wispy clouds. The overall scene is peaceful and historic.

Static Word Vectors

Lecture 04

Sources Lecture and Exercise

- Embeddings (Word2Vec)
 - <https://youtu.be/rmVRLeJRkl4?feature=shared&t=854> (to 39:59)
- Details of Word2Vec
 - <https://youtu.be/gqaHkPEZAew?feature=shared&t=826> (to 22:55)

Extra:

- Vector Representations of Words (before embeddings)
 - <https://youtu.be/ladnEWOntEM>

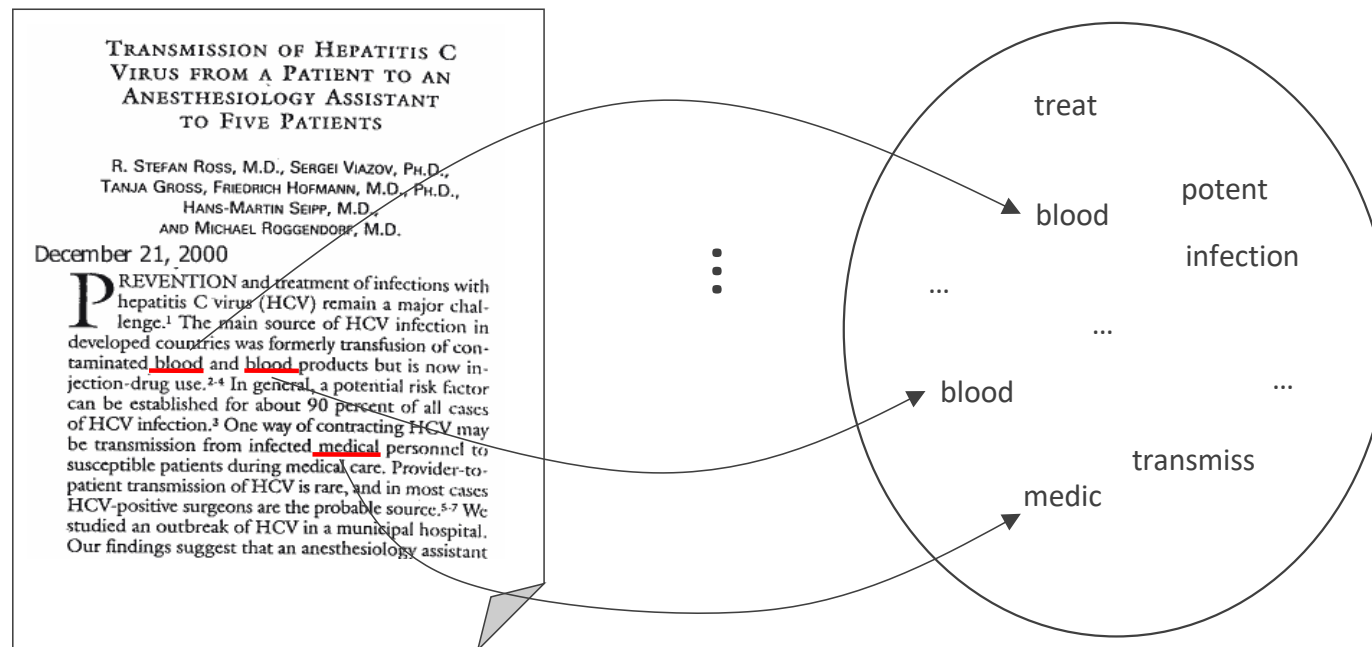
Flipped Classroom Questions

- What is “distributional semantics” about?
- Name a distributional representation of words that is not learned by machine learning.
- What is the input and output dimensions of the word2vec model?
- What is a softmax?

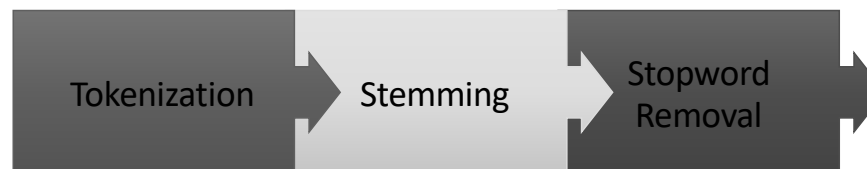
Words and Symbols



Recap: Represent Text by a set of symbols (= words/tokens)



typical full-text indexing pipeline



Recap: Bag-of-words Representation

- Sum of the word appearances per document
- Ignore the order of words

- Example:

Vocabulary = (Hotel, Motel, is, a, this)

Hotel Hotel = [2 0 0 0 0]

this is a Hotel = [1 0 1 1 1]

this is a Motel = [0 1 1 1 1]

is this a Hotel = [1 0 1 1 1]

- Here, we use *term frequency*, i.e., number of word occurrences in a document, as the weight of each word

Recap: Term-Document Matrix

	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7	Document 8
Term(s) 1	10	0	1	0	0	0	0	2
Term(s) 2	0	2	0	0	0	18	0	2
Term(s) 3	0	0	0	0	0	0	0	2
Term(s) 4	6	0	0	4	6	0	0	0
Term(s) 5	0	0	0	0	0	0	0	2
Term(s) 6	0	0	1	0	0	1	0	0
Term(s) 7	0	1	8	0	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0	0

← Word Vector
(Passage Vector)

Document Vector

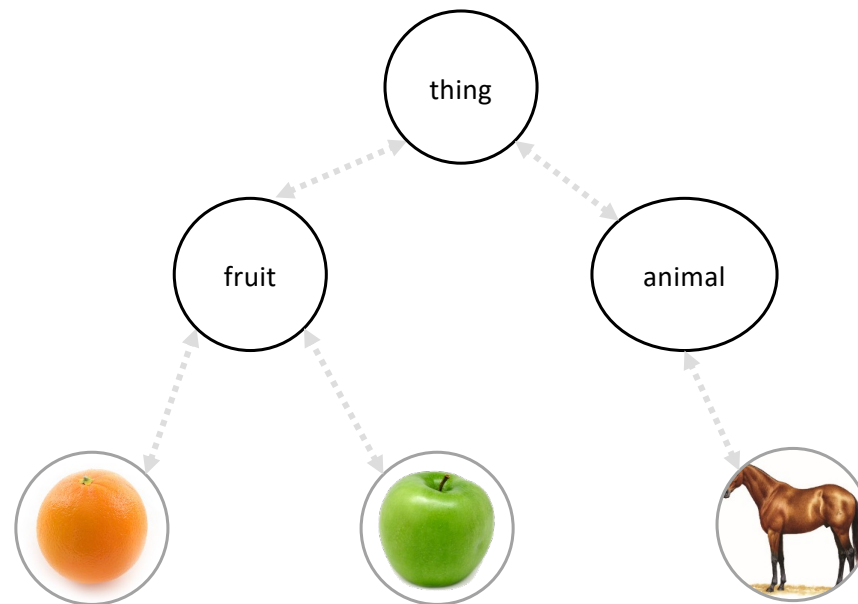
Measuring the Meaning of Words

Language is symbolic

- Text is a deliberately constructed discrete symbolic signaling system
 - Mostly exchange of symbols
- Word meaning: signifying an idea or thing
- The meaning of a word can be made explicit by linking it to a shared conceptualization, like an ontology or taxonomy

Measuring Meaning - Taxonomies

- Simple Benchmark: Semantic Similarity
- Simple semantic similarity: Shortest Path in Taxonomy



Drawback of Taxonomic representations

- The graph requires effort to construct and maintain
 - A shared conceptualization still is subjective
 - Similarity calculations are not robust
 - But without the attached graph the symbol is meaningless:
-
- | | | |
|------------|---|------------------------|
| Vocabulary | = | (Apple, Orange, Fruit) |
| Apple | = | [1 0 0] |
| Orange | = | [0 1 0] |
| Fruit | = | [0 0 1] |

One-HOT Encoding

- Index in a dictionary of words
- Apple $<[1 \ 0 \ 0]^T$, Orange $[0 \ 1 \ 0]^T > = 0$
- => obvious issue: Synonyms have 0 similarity (difficult to do calculations due to sparse representation)
- Sparse because vector is of size of the vocabulary
- Also, brains don't seem to work in a symbolic way, they are continuous systems

Measuring Meaning – String Distance

Name	Example
<u>Hamming distance</u>	"karolin" and "kathrin" is 3.
<u>Levenshtein distance</u>	kitten and sitting have a distance of 3: 1. kitten → sitten (substitution of "s" for "k") 2. sitten → sittin (substitution of "i" for "e") 3. sittin → sitting (insertion of "g" at the end).
<u>Most frequent k characters</u>	MostFreqKeySimilarity('research', 'seeking', 2) = 2

https://en.wikipedia.org/wiki/String_metric

Measuring Meaning – Manual Features

- Capitalized
- Contains a hyphen, numbers,...
- Frequency of the word in a corpus
- Parts of speech:

<https://www.cis.uni-muenchen.de/~schmid/tools/TreeTagger/data/Penn-Treebank-Tagset.pdf>

POS TAG	DESCRIPTION	EXAMPLE
JJ	adjective	green
JJR	adjective, comparative	greener
JJS	adjective, superlative	greenest
RB	adverb	however, usually, naturally, here, good
RBR	adverb, comparative	better
RBS	adverb, superlative	best
NN	common noun	table
NNS	noun plural	tables
NNP	proper noun	John
NNPS	plural proper noun	Vikings
VB	verb base form	take
VBD	verb past	took
VBG	gerund	taking
VBN	past participle	taken
VBP	verb, present, non-3d	take
VBZ	verb present, 3d person	takes
FW	foreign word	d'hoevre

Distributional Semantics of Words



Answer those multiple choice questions:

The word in question is being guessed correctly if

- the word is described in logically consistent statements.
- other words associated with the word in question are being used.
- the structures used to describe the word are grammatically correct.

Observation

- Meaning emerges from the relation to other symbols
- The meaning of a symbol can be conveyed, by the distribution of its neighbors as mentioned in language
- Horse := hoof, pony, cowboy, ...
- *“You shall know a word by the company it keeps”*
(J. R. Firth 1957:11)

Distributional semantics models (DSM)

- Intuitive approach: JoBimText (Biemann and Riedl, 2013)
- lexical items $j \in \mathcal{J}$ are represented by their p most salient contexts B_j (e.g., the surrounding words)
- Histogram of cooccurrence of words
- Similarity: $sim(j_k, j_l) = |\{x | x \in B_{j_k} \& x \in B_{j_l}\}|$

Graph-Based DSM

- Still have a graph of related explicit symbols
- Relatedness is calculated, not manually defined
- A symbol's meaning is represented by its p most frequent co-occurring symbols
- Word similarity is not directly computable for dissimilar pairs (still a discrete and sparse representation)

Dense Vector-based DSM

- Use an algebraic vector representation
 - Continuous values
 - Dense, fixed length, low-dimensional (100 – 1000)
 - Latent / hidden
- aka “embedding”
- *Adjust* the numbers so that each vector is good in predicting words in the context:
- Apple $<[1 \ 1]^T$, Orange $[1 \ 0.9]> = 1.9$

Words in Vector Space



Static Word Embeddings with W2V

aka “Type Embeddings”

Word Window

- Center word vs context word

I loved when I was a child to go but on a ranch...

$$p(w_{t-j}|w_t) \quad w_t \quad p(w_{t+j}|w_t)$$

Word2Vec Learning

1. Sample:

Select a word pair $(w_{t\pm j}|w_t)$ from corpus

2. Predict:

Predict the probability of this pair, given the current embeddings

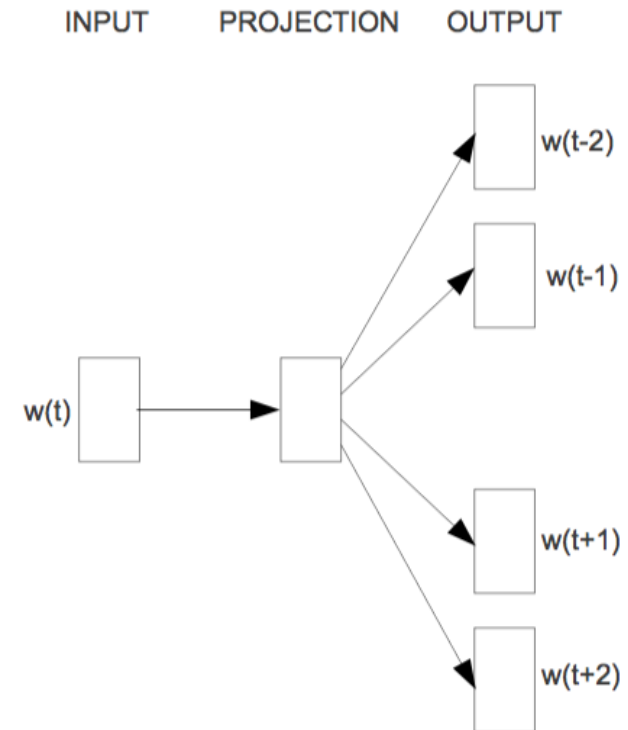
3. Optimize:

Change the embeddings, so that the predicted probability gets higher for your sampled word pair

Word2Vec Learning

1. Sample

2. Predict: $p(o|c) = \frac{\exp(u_o^T v_c)}{\sum_{w=1}^v \exp(u_w^T v_c)}$



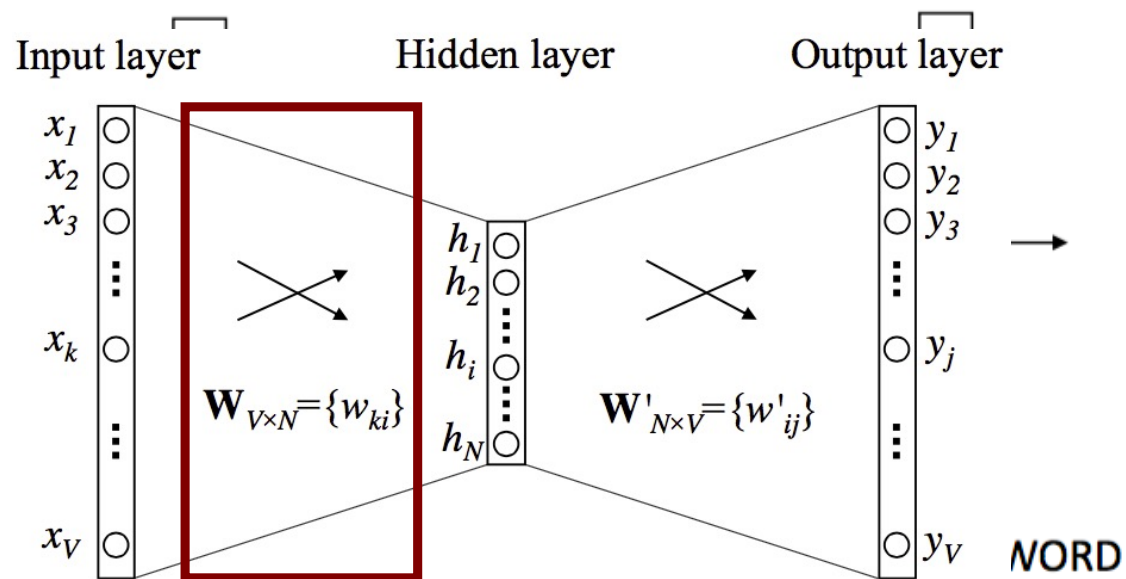
Word2Vec Learning

1. Sample:

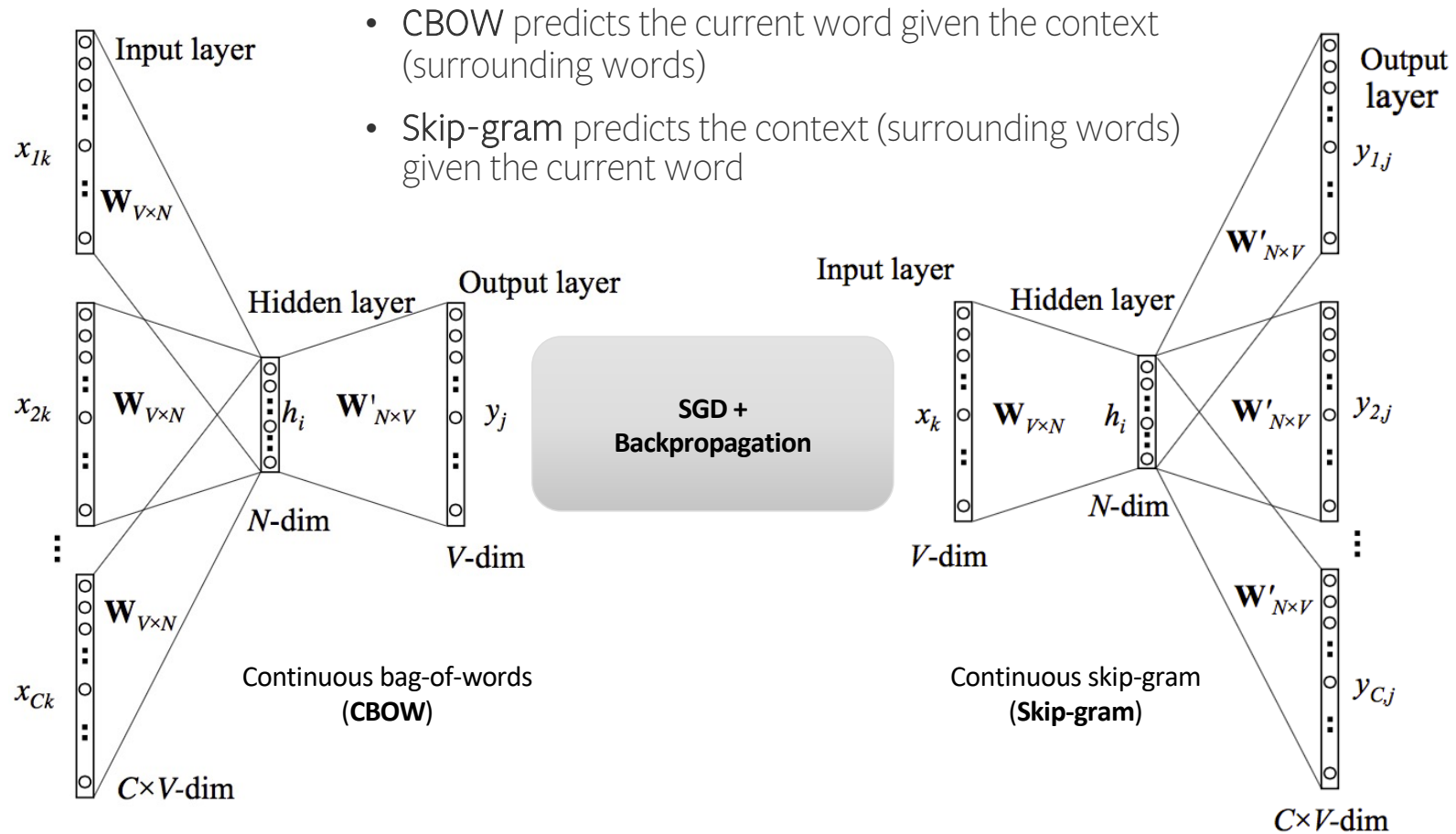
2. Predict:

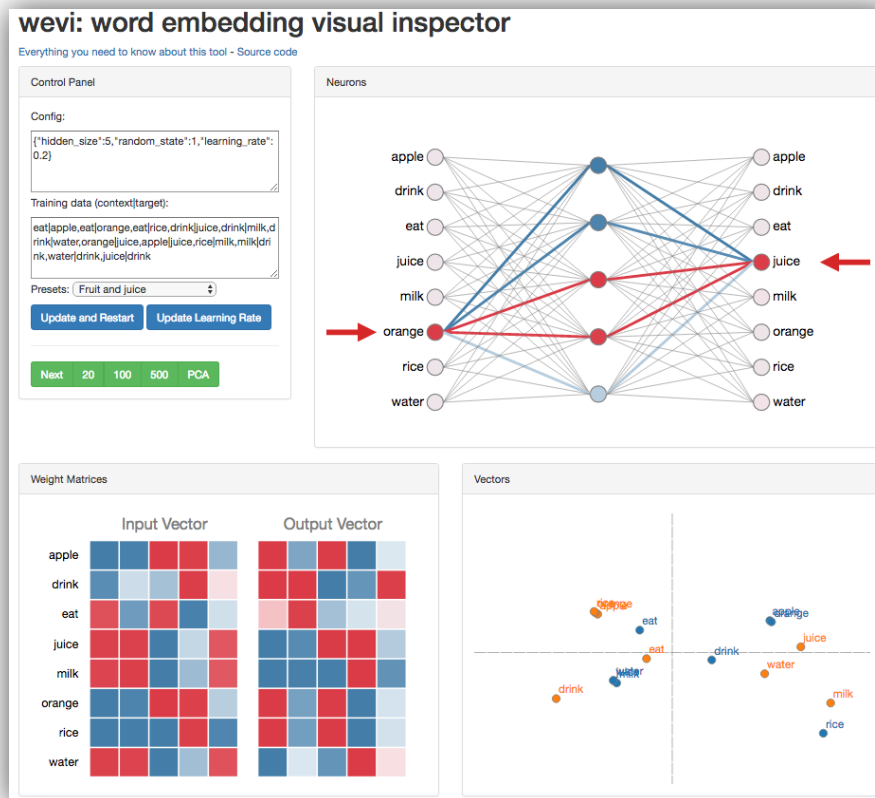
3. Optimize: $\max p(o|c) = \min \left(-\log \frac{\exp(u_o^T v_c)}{\sum_{w=1}^v \exp(u_w^T v_c)} \right)$

Neural Network perspective



Word2vec: CBOW and Skip-gram





<https://ronxin.github.io/wevi/>

Pros and Cons basic Word Embeddings

- Unsupervised training
- Easy to keep up-to-date
- Less subjective
- Vector reps are efficient for computations
- Vector arithmetics
- Not good with homographs
- No explicit structure and explainability
- Reasoning is very rudimentary, not expressive and not reliable
- Vector dimensions are task dependent

“The future of language is about the development of language models...”¹

1

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